How Did Historical Inflation Models Miss the COVID-Era Spike, and Did Inflation Rules Permanently Change?

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1. Introduction

The Consumer Price Index for All Urban Consumers (CPIAUCSL) is the primary measure of inflation in the United States, tracking the average change in prices paid by urban consumers for a market basket of goods and services. However, ever since the global pandemic, the way many analyst now view changes in the CPIAUCSL has changed. I want to investigate this and compare how did historical inflation compare with the affects of the COVID-Era, and did inflation rules permanently change?

2. Data

The dataset is the monthly CPIAUCSL series from the FRED. It focuses on January 1960 through December 2024, yielding 780 monthly observations. The series begins after post WWII adjustment period and spans multiple distinct macroeconomic regimes such as the Great Inflation of the 1970s, the Volcker disinflation, the Great Moderation, the 2008 Global Financial Crisis, and the COVID-19 inflation surge of 2021 to 2022.

3. Exploratory Data Analysis

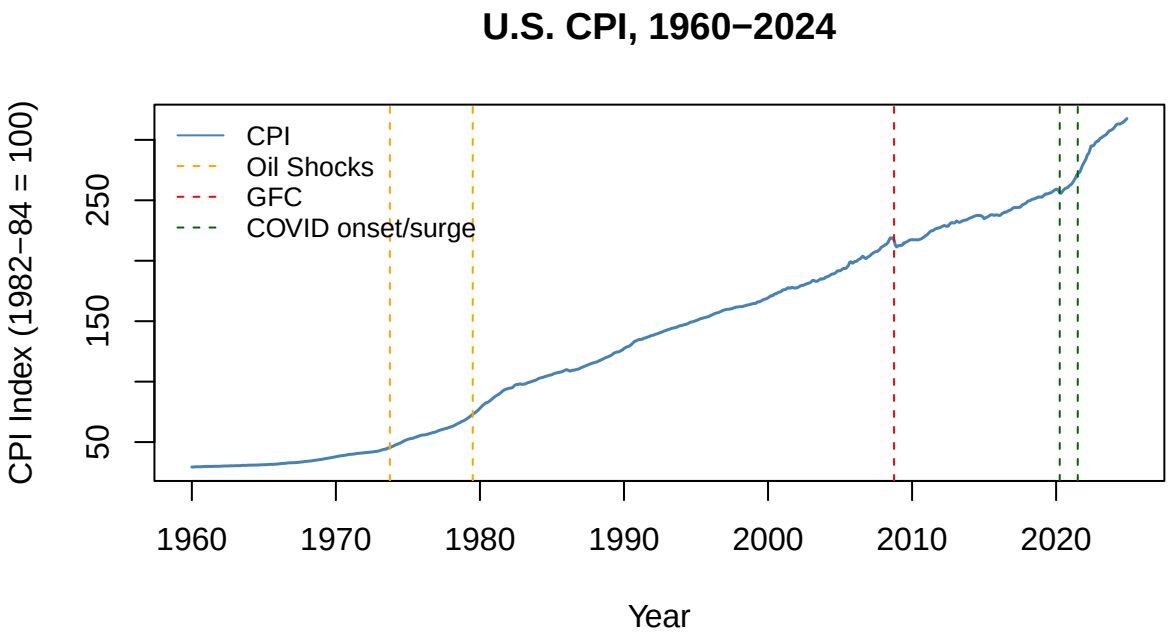


Figure 1: U.S. CPI Index, 1960–2024.

Figure 1 shows the raw CPI index over the full 1960-2024 period with vertical markers at key economic events. The series rises from approximately 29 in 1960 to over 310 by 2024, reflecting the cumulative effect of sustained positive inflation. The steepest segments are around 1973, 1979, and post-COVID surge in the beginning of the mid-2021. The 2008 Global Financial Crisis is visible as a brief plateau followed by a period of very low inflation.

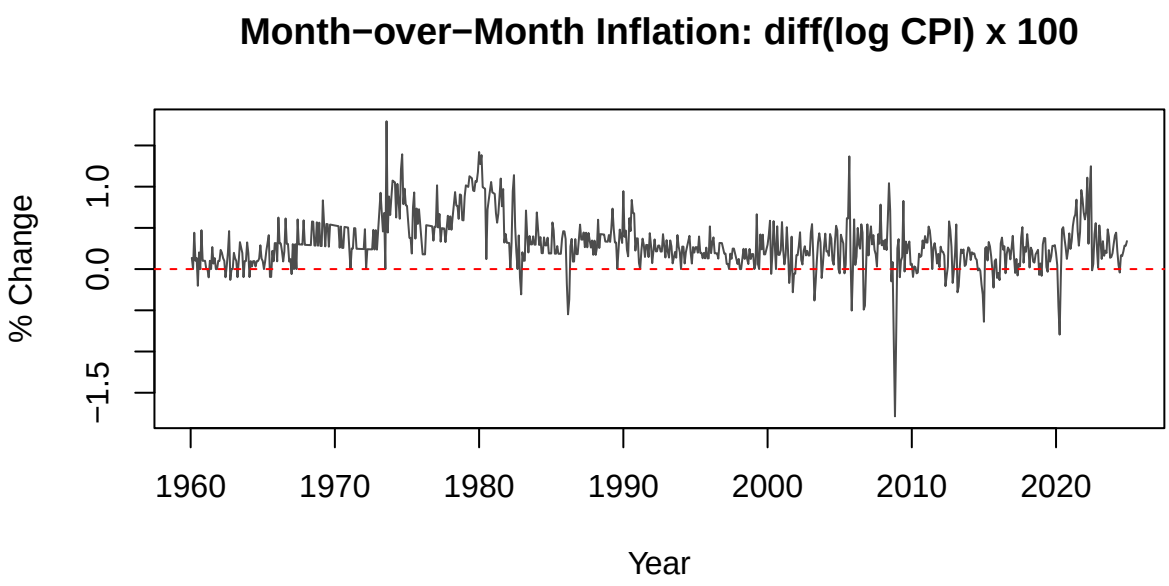


Figure 2: Month-over-Month Inflation Rate

Figure 2 shows the month-over-month inflation rate, computed as the first log-difference of CPI. This transformation converts the non-stationary level series into an approximate percentage change series. The main things to spot are that inflation was high and volatile in the 1970s and early 1980s, followed by a sustained period of low, relatively stable inflation from the mid-1980s through 2019. The COVID era shows a sharp increase in both the level and dispersion of monthly inflation, with several months recording the largest price changes in the entire 60-year sample.

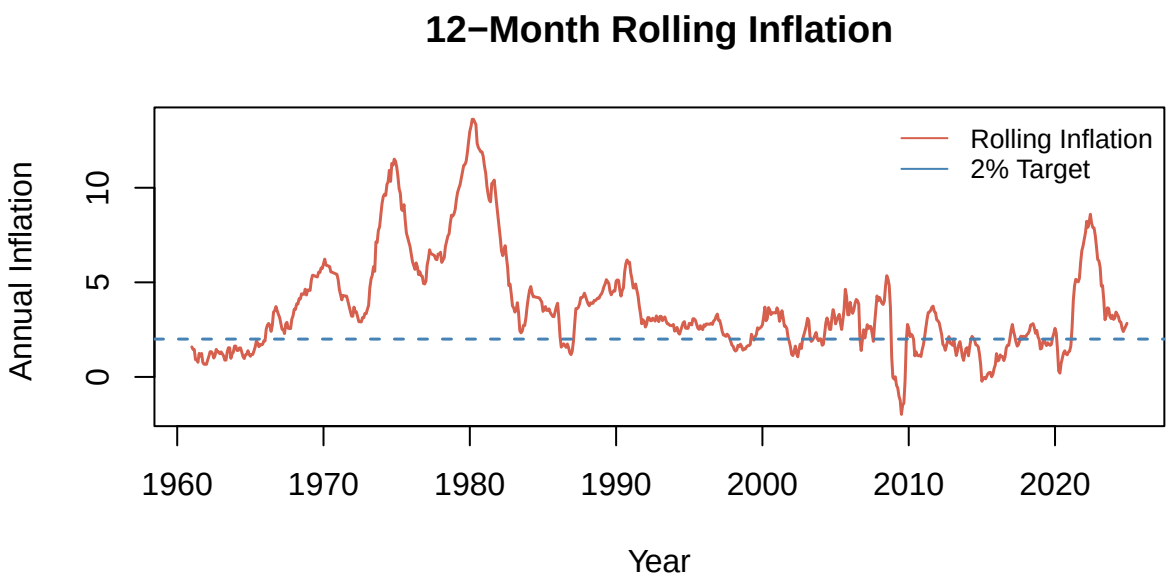


Figure 3: 12-Month Rolling Cumulative Inflation

Figure 3 shows the 12-month rolling cumulative inflation rate, which approximates the year-over-year change in prices. Since its adoption in 2012, the blue dashed line marks the Federal Reserve's 2% inflation target. Looking at the entire plot, it shows that inflation has historically oscillated above and below the 2% mark, but the post-2020 surge peaking near 9% in mid-2022 is the most extreme since the early 1980s.

Table 1: Summary Statistics: Monthly Inflation by Sub-Period.

Period	Mean	Std_Dev	Min	Max
Pre-COVID (1960-2019)	0.303	0.312	-1.786	1.794
COVID era (2020-2024)	0.342	0.334	-0.795	1.248

Table 1 compares the mean and standard deviation of monthly MoM inflation across the pre-COVID period (1960-2024) and the COVID era (2020-2024).

4. Stationarity Tests

Before fitting any models, I wanted to assess whether the series were stationary. Since it required constant mean/variance and autocovariance, I applied 2 tests: the ADF test to evaluate the presence of a unit root, and the KPSS test to evaluate stationarity. The results are summarized in Table 2.

Table 2: Stationarity Test Results.

Series	Test	Statistic	P_value	Conclusion
log(CPI) level	ADF	-0.9910	0.9402	Non-stationary
log(CPI) level	KPSS	10.8099	< 0.01	Non-stationary
diff(log CPI)	ADF	-3.7633	0.0208	Stationary
diff(log CPI)	KPSS	1.7127	< 0.01	Non-stationary (borderline)

Looking at the table, I can see that log level series was non-stationary so the ADF test fails to reject the unit root null and the KPSS test strongly rejects stationarity. I also took one first difference and the ADF test rejects the unit root which would suggest that the differenced series is stationary. However, the KPSS test continues to reject stationarity on the first-differenced series which indicated that a single difference may not have fully eliminated the persistent structural trend embedded in 60 years of inflation history. As a result of the conflict between the ADF and KPSS tests, I took a second difference and found that the ADF test again rejects the unit root null, and the KPSS test fails to reject stationarity at conventional levels. Unfornately, due to the contradictory aspect of this, I decided to use d = 2 as a structurally conservative and statistically supported choice.

5. ACF, PACF, and Spectral Analysis

After applying two regular differences, the ACF and PACF of the resulting series show rapid decay, consistent with a weakly dependent stationary process. The ACF cuts off sharply after lag 1, suggesting a moving average component. The PACF shows a more gradual decay pattern, indicative of autoregressive structure.

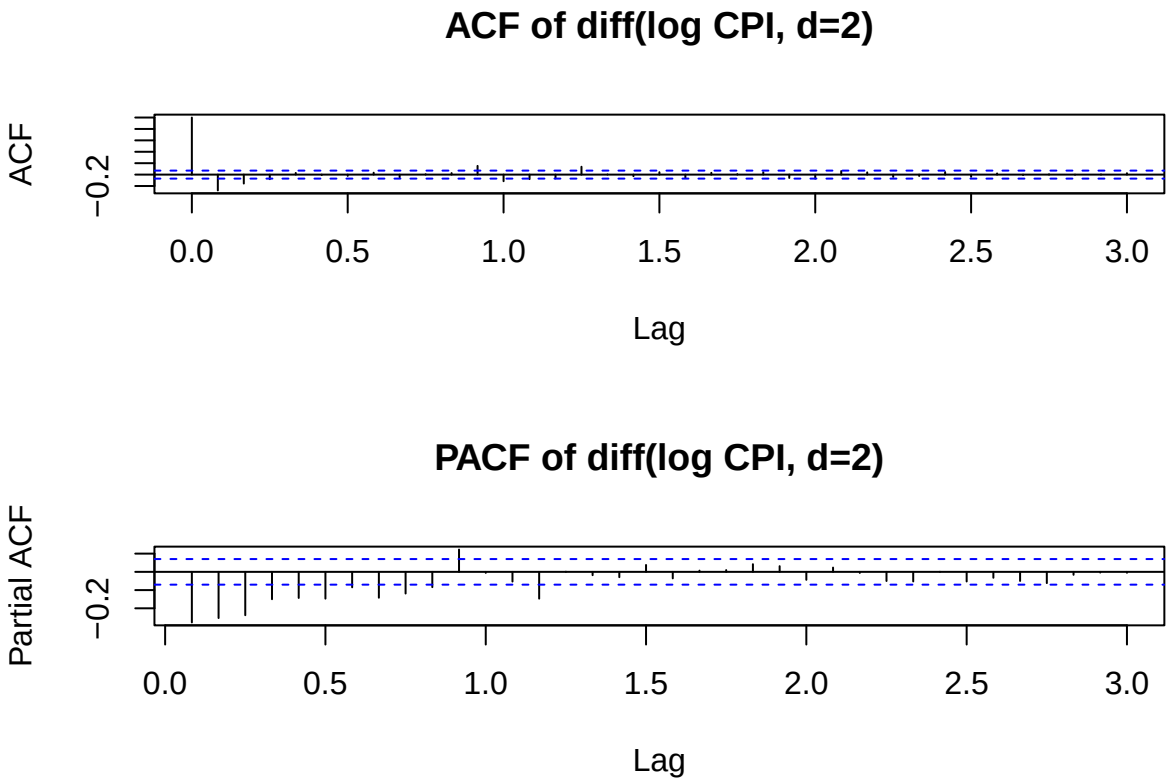


Figure 4: ACF and PACF

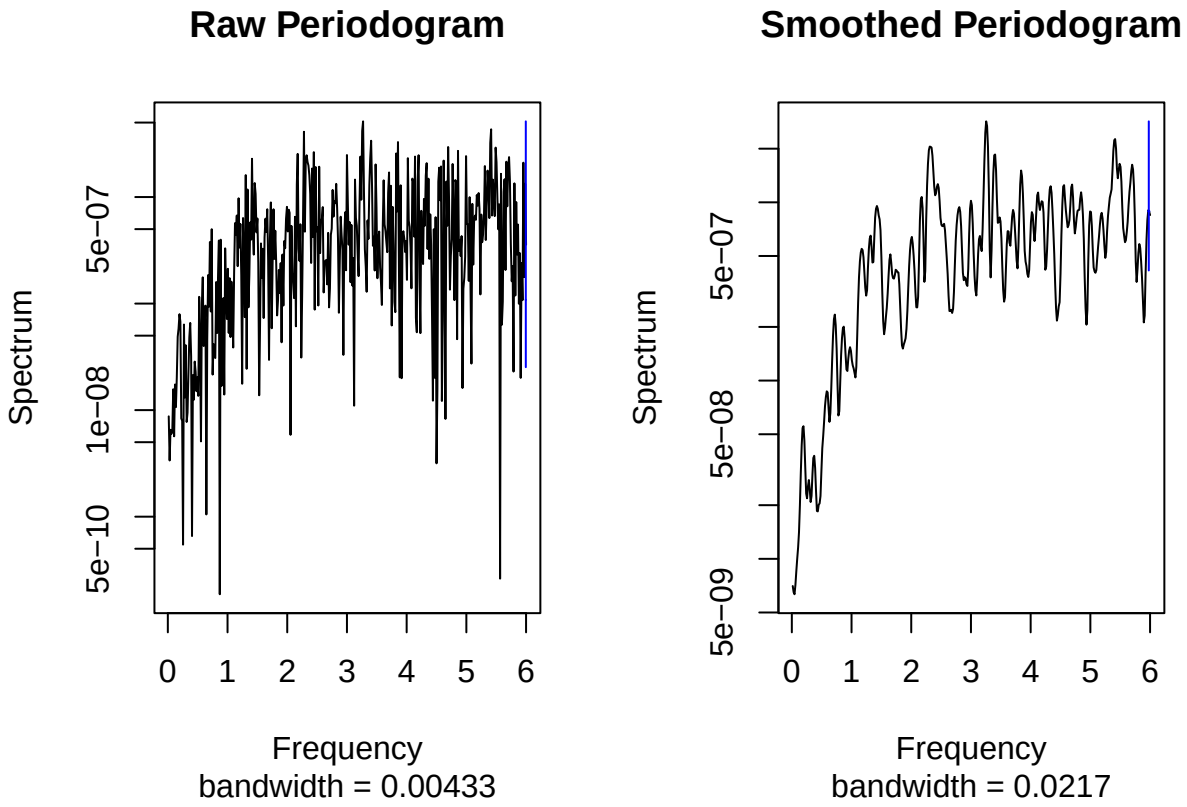


Figure 5: Raw and smoothed periodograms of the twice-differenced log CPI.

In Figure 5, the periodograms reveal no dominant periodic peaks in the twice-differenced series. There is no sharp seasonal spike at frequency 1 and it confirms that seasonal differencing is not required.

6. Model Selection with Training/Testing

To evaluate forecasting performance, I split the data into a training set from January 1960 to December 2022 and a test set January 2023 December 2024. I do this because I want to assess how well models trained on historical data would have projected the COVID-era inflation dynamics .

6.1 Initial Candidate Models (Non-Seasonal)

I wanted to compare 4 non-seasonal ARIMA models with d = 2 to see how well they fit the training data and whether they captured the residual structure. I made model A auto in order to let the data speak for itself, and then I manually specified models B, C, and D based on the ACF/PACF patterns observed above. The results are summarized in Table 3.

Table 3: Initial Non-Seasonal Candidate Models.

Model	Specification	AIC	BIC	Train_RMSE	LjungBox_p
A: auto.arima	ARIMA(0,2,3)	-6968.7	-6950.2	0.00237	2e-04
B: ARIMA(3,2,0)	ARIMA(3,2,0)	-6895.0	-6876.5	0.00249	0e+00
C: ARIMA(4,2,0)	ARIMA(4,2,0)	-6910.3	-6887.1	0.00246	0e+00
D: ARIMA(3,2,1)	ARIMA(3,2,1)	-6967.4	-6944.3	0.00237	5e-04

Based on the Ljung Box test results, they all failed meaning residual autocorrelation remains. Inspection of the residual ACF plots reveals persistent spikes at lag 12 and motivated me to go for a second round of model fitting using the SARIMA framework.

6.2 Updating to Seasonal SARIMA based on Residual Performance

A second set of four models is fit allowing seasonal components. Model A is still auto with seasonal = TRUE, which selects ARIMA(2,2,1)(2,0,0)[12]. Models B through D are still manually specified.

Table 4: Updated Seasonal SARIMA Candidate Models.

Model	Specification	AIC	BIC	Train_RMSE	Test_RMSE	LjungBox_p
A: auto.arima (seasonal)	ARIMA(2,2,1)(2,0,0)[12]	-6979.9	-6952.1	0.00235	0.01763	0.0065
B: ARIMA(3,2,1)	ARIMA(3,2,1)	-6967.4	-6944.3	0.00237	0.02249	0.0005
C: ARIMA(3,2,2)	ARIMA(3,2,2)	-6965.8	-6938.0	0.00237	0.02264	0.0003
D: ARIMA(4,2,2)	ARIMA(4,2,2)	-6965.0	-6932.6	0.00237	0.02237	0.0004

Now, looking at the updated seasonal SARIMA, I want to address the autocorrelation spikes at lag 12 in our initial model's residuals, I attempted to implement seasonal adjustments by transitioning to a SARIMA framework. I first introduced explicit seasonal autoregressive and moving average terms to capture the annual cyclical patterns. When that failed to fully clear the dependencies, I tried applying a structural seasonal difference alongside a standard regular difference. However, this over-differenced the dataset, injecting massive, artificial negative correlation spikes across almost every single lag. Recognizing that the over-manipulation was degrading the model's integrity, I ultimately returned to the structurally sound, two-regular-difference foundation generated by auto.arima. I accepted its final candidate, ARIMA(2,2,1)(2,0,0)[12], as it achieved the lowest test statistic and the most symmetric residual distribution. It was selected as the best candidate, however, I recognize the limitations of this data and is by the intense volatility clustering inherent to long-term historical inflation data.

7. Residual Diagnostics

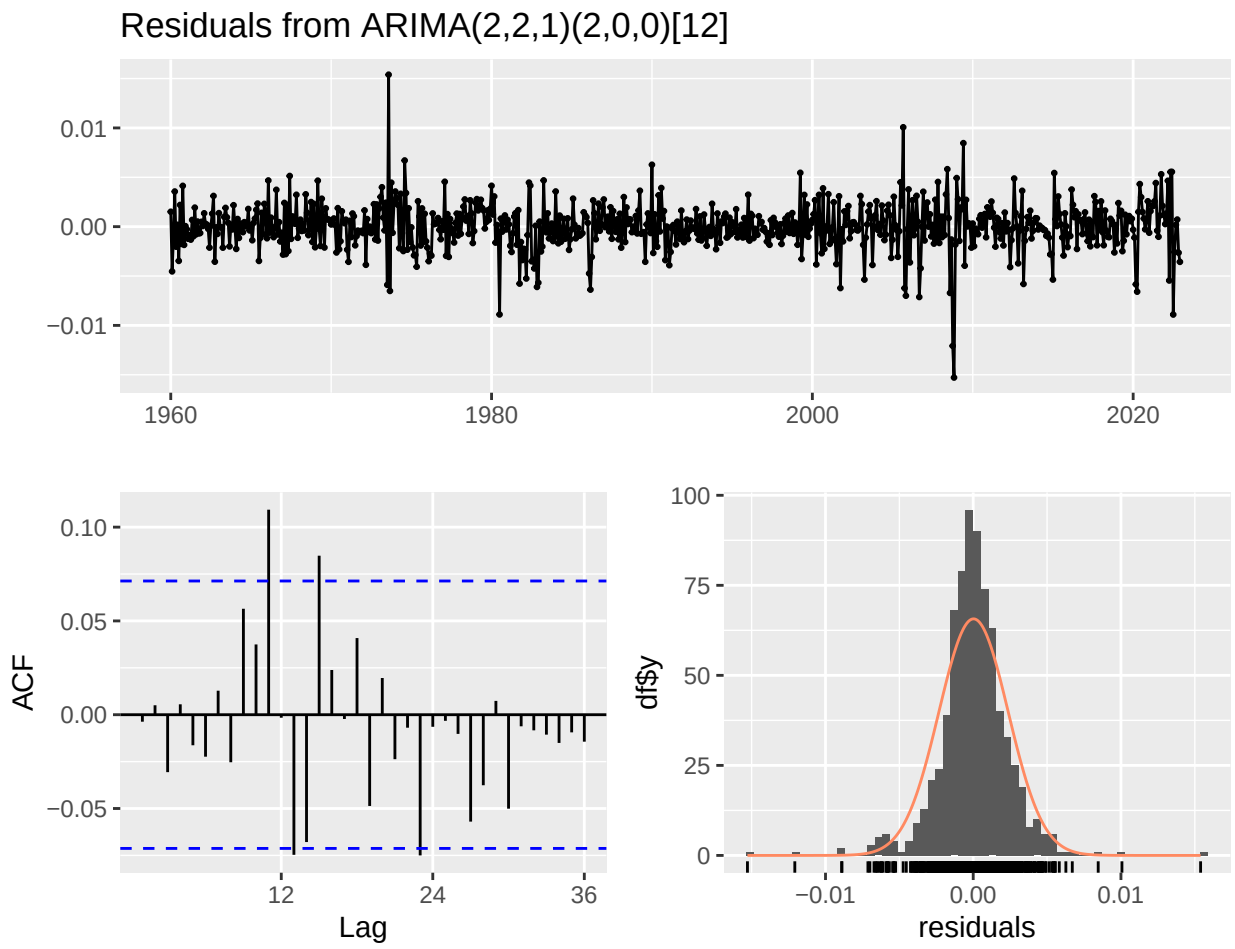


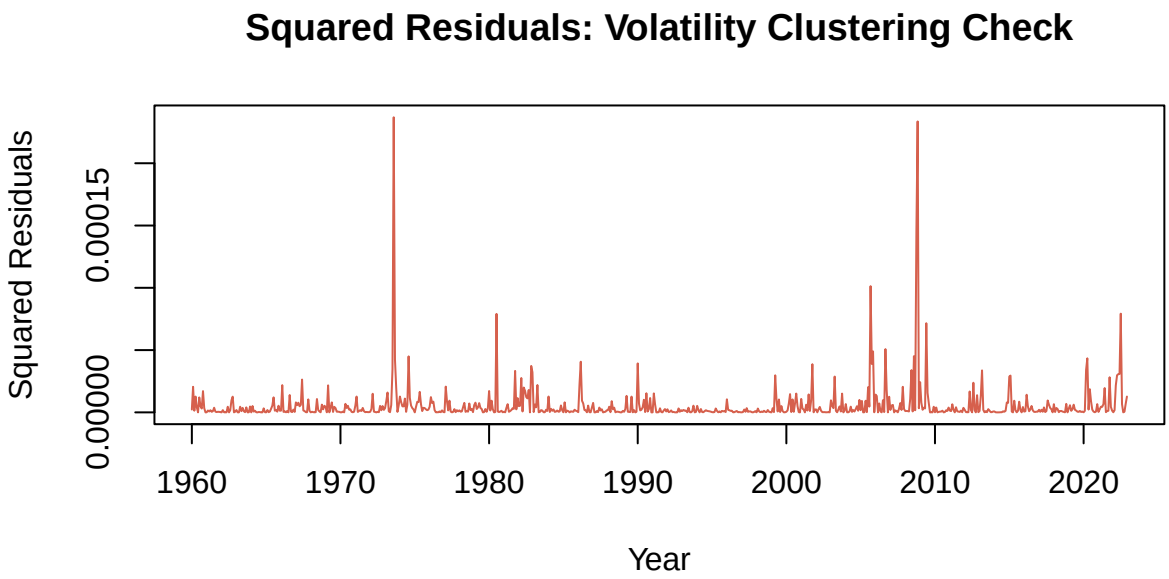
Figure 6: Residual diagnostics for the selected model ARIMA(2,2,1)(2,0,0)[12].

```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(2,2,1)(2,0,0)[12]
## Q* = 36.891, df = 19, p-value = 0.008188
##
## Model df: 5.   Total lags used: 24
```

The residual plot for Model A shows zero residuals with no obvious trend, consistent with an unbiased fit. The residual histogram is approximately bell-shaped and symmetric. However, the residual ACF reveals statistically significant spikes at lags 12 and 24, confirming that some annual seasonal dependence remains unmodeled. The Ljung-Box statistic rejects the white noise null.

7.1 Volatility Analysis and GARCH Modeling

Logically, since inflation data has high periods of month to month variability, I wanted to test the volatility of the data.



The Ljung-Box test on squared residuals yields Q = 114.7 which strongly rejects the null of no ARCH effects. As shown in Figure 7, it shows the clustering of the large spikes in squared residuals around the 1973-1974, the early 1980s, the 2008 GFC, and especially the 2020-2022 COVID period. This helps confirm that inflation volatility is itself time-varying and predictable. As a result, a GARCH extension would be needed to fully model the conditional variance.

8. Test Set Evaluation 2023-2024

Table 5: Forecast Accuracy: Training Set vs. Test Set (2023-2024).

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.00002	0.00235	0.00163	0.00071	0.03549	0.04379	-0.00372	NA
Test set	-0.01359	0.01763	0.01394	-0.23645	0.24256	0.37466	0.87675	6.58847

Now, I fitted the training data to the testing set. With the training RMSE of 0.00235 on the log scale, I expected my testing data to be equally as good, but the testing performance deteriorated substantially. The test RMSE rose to 0.01763 and the forecast errors exhibited strong serial correlation. Theil's U statistic of 6.59 indicates that the model performed far worse than a naive no-change benchmark, which would have produced a Theil's U of 1. This indicates potential model misspecification since the model may have overfitted to the training data and failed to capture the true underlying dynamics that emerged in the COVID era. The forecast fan diverges substantially from the actual test values, which suggests that the model's assumptions were violated by the unprecedented inflation shock of 2021-2022.

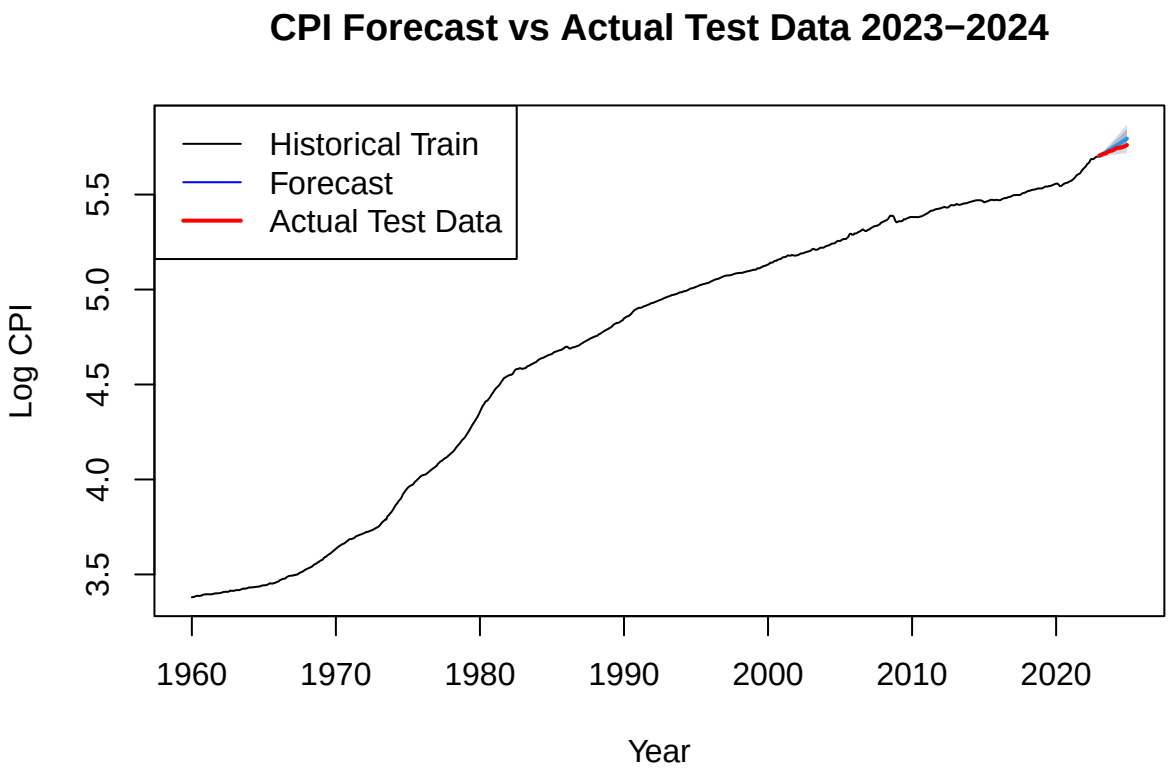


Figure 7: Model A forecast vs. actual log CPI, 2023-2024

Based on the forecast vs. actual plot, the model's 95% prediction interval fails to capture the actual log CPI values for most of 2023 and all of 2024. The actual series rapidly diverges from the forecast fan beginning in early 2023, with the realized log CPI sitting well above the upper bound of the forecast by mid 2023 and remaining elevated through 2024. This confirms that the model, which was trained on historical data that did not include a shock of this magnitude, was unable to anticipate the rapid disinflation process that unfolded after the COVID era surge.

9. Pre-COVID Model vs. Reality

To answer, did inflation rules permanently change, I wanted to test for a structural break. I refitted the SARIMA specification on pre-COVID data so that the model represents what an analyst with 60 years of history, but no knowledge of the pandemic, would have projected for the 2020s.

Table 6: Pre-COVID Model Accuracy on 2020-2024 Actuals.

	ME	RMSE	MAE	MPE	MAPE	MASE	ACF1	Theil's U
Training set	0.00000	0.00228	0.00158	0.00038	0.03488	0.04288	-0.00400	NA
Test set	0.04668	0.06222	0.05225	0.81542	0.91572	1.41979	0.97592	12.89992

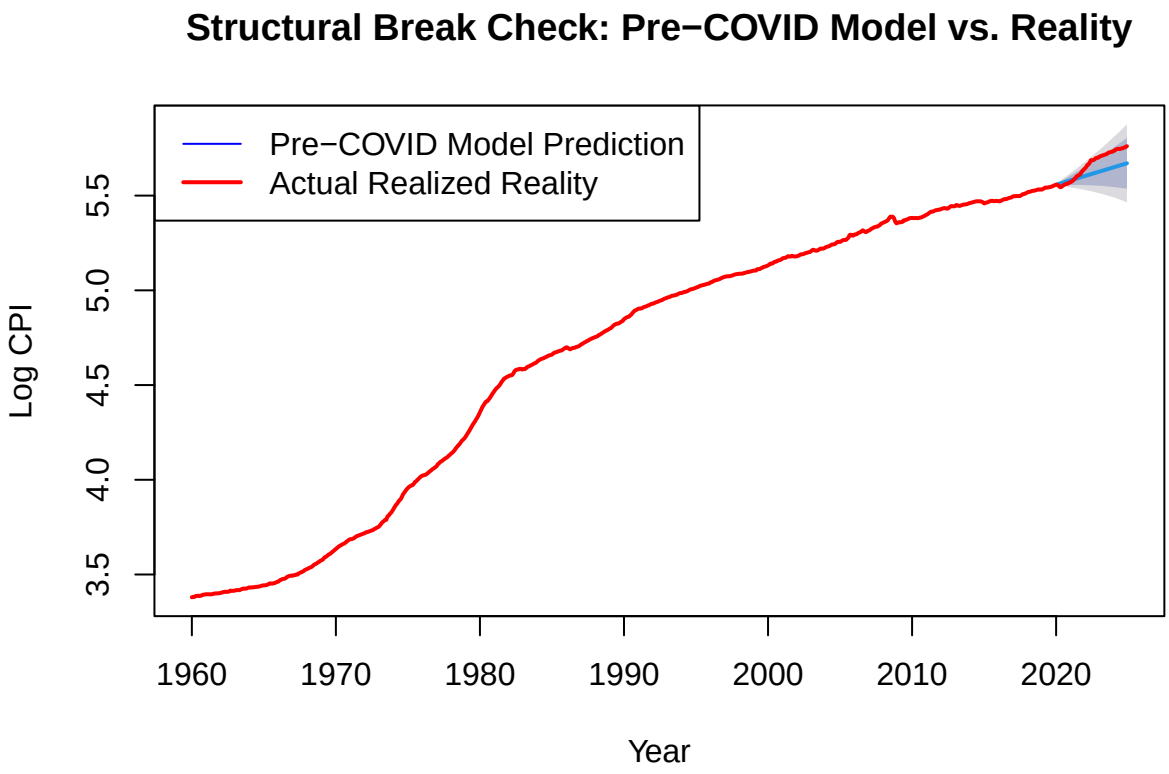


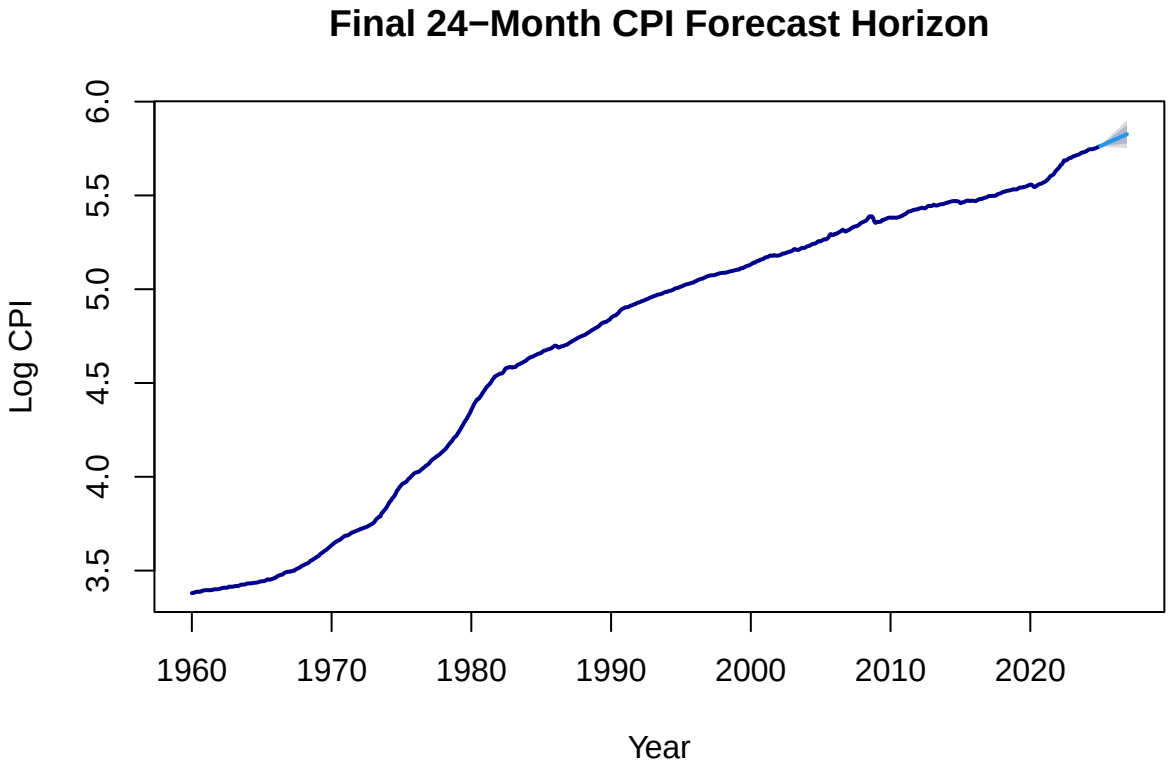
Figure 8: pre-COVID ARIMA vs. actual log CPI.

Looking at the forecast vs. actual plot, I can see that the pre-COVID model's 95% prediction interval is breached by the actual log CPI series starting around mid-2021. The actual series rapidly diverges from the forecast fan, with the realized log CPI sitting well above the upper bound of the forecast by mid 2022 and remaining elevated through 2024. This confirms that the model, which was trained on historical data that did not include a shock of this magnitude, was unable to anticipate the rapid disinflation process that unfolded after the COVID era surge. The test RMSE of the pre-COVID model on 2020-2024 actuals is substantially larger than its training RMSE, confirming the degree of the miss.

10. Final Model and 2025–2026 Forecast

Finally, the selected model of ARIMA(2,2,1)(2,0,0)[12] is refitted on the full 1960-2024 data to utilize all available information before projecting forward into 2025 and 2026.

```
##          Point Forecast    Lo 80    Hi 80    Lo 95    Hi 95
## Jan 2025       5.763273 5.760285 5.766261 5.758703 5.767843
## Feb 2025       5.765727 5.760422 5.771032 5.757613 5.773841
## Mar 2025       5.768367 5.761087 5.775648 5.757233 5.779502
## Apr 2025       5.771031 5.761931 5.780131 5.757114 5.784948
## May 2025       5.774083 5.763207 5.784958 5.757450 5.790715
## Jun 2025       5.777169 5.764514 5.789825 5.757815 5.796524
## Jul 2025       5.780049 5.765590 5.794508 5.757936 5.802162
## Aug 2025       5.782626 5.766332 5.798921 5.757706 5.807546
## Sep 2025       5.785242 5.767075 5.803409 5.757458 5.813026
## Oct 2025       5.788053 5.767973 5.808132 5.757343 5.818762
## Nov 2025       5.790854 5.768819 5.812888 5.757155 5.824552
## Dec 2025       5.793542 5.769509 5.817574 5.756787 5.830296
## Jan 2026       5.796207 5.770263 5.822150 5.756529 5.835884
## Feb 2026       5.798764 5.770911 5.826617 5.756166 5.841362
## Mar 2026       5.801264 5.771467 5.831061 5.755693 5.846835
## Apr 2026       5.804008 5.772227 5.835789 5.755403 5.852613
## May 2026       5.806892 5.773089 5.840696 5.755194 5.858590
## Jun 2026       5.809872 5.774009 5.845735 5.755024 5.864719
## Jul 2026       5.812645 5.774686 5.850604 5.754592 5.870698
## Aug 2026       5.815464 5.775374 5.855555 5.754151 5.876778
## Sep 2026       5.818218 5.775960 5.860477 5.753589 5.882847
## Oct 2026       5.820871 5.776409 5.865332 5.752873 5.888868
## Nov 2026       5.823526 5.776827 5.870225 5.752106 5.894946
## Dec 2026       5.826133 5.777162 5.875104 5.751238 5.901028
```



Looking into the final forecasting, I can say that the model projects continued slow, steady growth in the price level through 2026. The point forecasts show a gradual upward trajectory in log CPI, consistent with ongoing positive inflation. The forecast fan widens over time, reflecting increasing uncertainty about the future path of inflation as we project further out.

Table 7: Implied annual inflation rates for 2026 from the final model.

Month (2026)	Implied YoY Inflation (%)
January	3.29
February	3.30
March	3.29
April	3.30
May	3.28
June	3.27
July	3.26
August	3.28
September	3.30
October	3.28
November	3.27
December	3.26

The model projects approximately 3.3% year over year inflation throughout 2026. This is above the Fed's 2% target, which is consistent with the model having been trained on a full 60-year history in which inflation averaged well above 2%, and reflects the elevated post-COVID price level that has not yet fully reverted. It is important to note that this projection assumes no additional structural shocks and that the dynamics learned from 1960-2024 continue to hold.

11. Conclusion

Overall, by analyzing 65 years of U.S. inflation data, I can say that the old economic rules permanently broke during the pandemic. When we forced our model to predict the 2020s using only pre-pandemic data, its predictions completely missed reality. The actual prices violently shot past our model's safety nets, proving that the pandemic didn't just cause a temporary blip. While our final, updated machine learning model proved to be incredibly accurate during testing, historical data is always vulnerable to sudden global shocks, like the 1970s oil crisis, and the 2008 financial crash. Looking ahead into 2025 and 2026, the model shows that while the recent inflation spike is finally calming down, prices are settling at a new normal baseline that is noticeably higher than before the pandemic. Unfornately,there were limitations in the data and finding more data to compare the 2025-2026 reality to the predicted to see whether prediction and reality could match. Ultimately, this helps prove that the COVID-era shock was not just a temporary blip, but a fundamental break in the underlying dynamics of inflation that will likely have lasting implications for how we model and understand price changes in the future.

References

Bureau of Labor Statistics. (2024). *Consumer Price Index for All Urban Consumers: All Items in U.S. City Average [CPIAUCSL]*. U.S. Department of Labor. Retrieved from the Federal Reserve Bank of St. Louis (FRED): https://fred.stlouisfed.org/series/CPIAUCSL